

Government Polytechnic, Dharashiv.

Second Formative Assessment of Theory (FA-TH) Test AY-2024-25

Program Name: Computer Engineering Program (CO)

Semester : Third (CO3K)

Course Title : Data Structures Using C - 313301

Marks : 30 Duration: 1.5 Hour Date : 12/11/2024

CO N	Course Outcome(CO)
CO1	Perform Basic Operations on Array
CO2	Apply different searching and sorting methods.
CO3	Implement basic operations on Linked List.
CO4	Perform operations on Stack using Array and Linked list Implementation.
CO5	Perform operations on Queue using Array and Linked list Implementation.
CO6	Create and traverse Tree to solve Problems.

Instructions: Question 1 and 2 are compulsory.

Q1	Attempt any Five (2x5=10 Marks)	Marks	CO
a	List Application of Linked List?	2	3
b	With Diagram, Explain Stack?	2	4
c	Define : (i) General tree (ii) Binary tree	2	6
d	Differentiate between Linear Queue and Circular Queue?	2	5
e	With Diagram, Explain Circular Queue?	2	5
f	List any four applications of queue?	2	5
g	Explain In-degree and Out-degree w.r.t Tree?	2	6

Q2	Attempt any Five (4x5=20 Marks)	Marks	CO
a	Construct a binary search tree for following elements, 22, 27, 14, 31, 40, 43, 44, 10, 20, 35 Show each step of construction of BST.	4	6
b	Show the effect of PUSH and POP on stack contains 10, 20, 25, 15, 30 & 40 Show diagrammatically the effect of (i) PUSH (45) (ii) PUSH (50) (iii) POP (iv) PUSH (55)	4	4
c	Write Recursive algorithm for In-Order,Pre-Order Tree traversal?	4	6
d	Convert into Post fix form give stepwise procedure $A+B^{\wedge}C*(D/E) - F/G$	4	4
e	Explain Queue overflow and underflow conditions with example?	4	5
f	Write a program to implement a stack with push, pop and display operations.	4	4
g	Explain Concept of Recursion using Stack? Give Example	4	4

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